

ECE 1508: Reinforcement Learning

Chapter 1: Introduction

Ali Bereyhi

`ali.bereyhi@utoronto.ca`

Department of Electrical and Computer Engineering
University of Toronto

Fall 2026

Playing in RL Framework: *Policy*

As environment gets to state S_t , agent decides for an action

↳ We can describe this behavior via *policy function*

Policy

Policy $\pi(a|s)$ is a probability distribution over a conditional to s

$$\pi(a|s) = \Pr\{A_t = a | S_t = s\}$$

- + Why a probability distribution? Does agent always act *randomly*?!
- It can either act *randomly* or *deterministically*, and both these cases can be presented by this notation

Playing in RL Framework: *Probabilistic Policy*

Consider the set of possible actions

$$\mathbb{A} = \{a^1, \dots, a^M\}$$

Agents might behave *probabilistic*, e.g.,

$$\pi(a|s) = \begin{cases} 0.8 & a = a^1 \\ 0.2 & a = a^2 \\ 0 & \text{otherwise} \end{cases}$$

This means that in state s

- in 80% of cases it acts a^1
- in 20% of cases it acts a^2
- it never does any other action

Playing in RL Framework: *Deterministic Policy*

Agents might behave *deterministically*, e.g.,

$$\pi(a|s) = \begin{cases} 1 & a = a^1 \\ 0 & \text{otherwise} \end{cases}$$

This means that if the agent observes state s

- it always acts a^1*
- it never does any other action*

Example: Tic-Tac-Toe

0	1	2
3	4	5
6	7	8

X		

Tic-Tac-Toe. *Oponent* starts at *cell 0*

- State is then $S_1 = \{0\}$: *Policy* specifies *next move*

$$\pi(a|\{0\}) = \Pr\{A_1 = a | S_1 = \{0\}\} = \begin{cases} p_a & a = 1, \dots, 8 \\ 0 & a = 0 \end{cases}$$

for some p_a 's that add up to 1

Example: Tic-Tac-Toe

0	1	2
3	4	5
6	7	8

X	O	

We may play *deterministically*: we are *sure* what to do next

- We may always play $A_1 = 1$ after seeing *opponent playing zero*

$$\pi(a | \{0\}) = \begin{cases} 1 & a = 1 \\ 0 & a \neq 1 \end{cases}$$

Example: Tic-Tac-Toe

0	1	2
3	4	5
6	7	8

X		
O		

Or

- We may always play $A_1 = 3$ after seeing *opponent playing zero*

$$\pi(a | \{0\}) = \begin{cases} 1 & a = 3 \\ 0 & a \neq 3 \end{cases}$$

Example: Tic-Tac-Toe

0	1	2
3	4	5
6	7	8

X	O	
O	O	

Or we may play *probabilistic*

- We may play *any* next cell $A_1 \in \{1, 3, 4\}$

$$\pi(a | \{0\}) = \begin{cases} 1/3 & a \in \{1, 3, 4\} \\ 0 & a \notin \{1, 3, 4\} \end{cases}$$

Example: Tic-Tac-Toe

0	1	2
3	4	5
6	7	8

X	O	
X		

Say we acted $A_1 = 1$, and the *opponent plays 3* after our move

- *Policy* should tell us the *next move*

↳ State changes to $S_2 = \{0, 1, 3\}$ and our policy should say

$$\pi(a | \{0, 1, 3\}) = \Pr\{A_2 = a | S_2 = \{0, 1, 3\}\}$$

? How many distributions are presented by *the policy*?

↳ Think of it as a *table*

Playing in RL Framework: *Value Function*

Agent can use **its policy** and **rewarding function** to estimate *future return*

Value Function

Assume agent acts with policy $\pi(a|s)$ at **state** s ; then, value of **this state** is

$$v_{\pi}(s) = \mathbb{E}\{G_t | S_t = s\}$$

Because it depends on **policy** π

- + Why do we have index π ?
- Because this value depends on the **policy**

Playing in RL Framework: Value Function

Let's break it down: we recall that

Future return at time t

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

for some discount factor $0 \leq \gamma \leq 1$

Also, we have

Rewarding function is going to determine the next reward

$$R_{t+1} = \mathcal{R}(S_t, A_t)$$

and *transition function* determines the next state

$$S_{t+1} = \mathcal{P}(S_t, A_t)$$

Playing in RL Framework: *Value Function*

Value of *state* s is

$$v_{\pi}(s) = \mathbb{E} \{G_t | S_t = s\} = \mathbb{E} \{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s\}$$

At time t , we see *state* $S_t = s$

- *Policy* $\pi(a|s)$ specifies *next move* A_t
- *Rewarding function* specifies the *immediate reward*

$$R_{t+1} = \mathcal{R}(S_t, A_t)$$

- *Transition function* specifies the *next state*

$$S_{t+1} = \mathcal{P}(S_t, A_t)$$

Playing in RL Framework: Value Function

Value of *state* s is

$$v_{\pi}(s) = \mathbb{E}\{G_t | S_t = s\} = \mathbb{E}\{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s\}$$

Rewarding function specifies *distribution of* R_{t+1}

- For a given *state* s and *action* a , we get

$$p_{\ell}(s, a) = \Pr\{R_{t+1} = r^{\ell} | S_t = s, A_t = a\}$$

- We can find expected *immediate reward*

$$\begin{aligned} \mathbb{E}_{\pi}\{R_{t+1} | S_t = s\} &= \sum_{m=1}^M \mathbb{E}\{R_{t+1} | S_t = s, A_t = a^m\} \pi(a^m | s) \\ &= \sum_{\ell=1}^L \sum_{m=1}^M p_{\ell}(s, a^m) \pi(a^m | s) r^{\ell} \end{aligned}$$

Playing in RL Framework: Value Function

Note that distribution of R_{t+1} can change, if we change policy

Example. Say rewarding function says $\mathcal{R}(s, a^1) = 0$ and $\mathcal{R}(s, a^2) = 1$

$$\pi^1(a|s) = \begin{cases} 1 & a = a^1 \\ 0 & a \neq a^1 \end{cases} \quad \pi^2(a|s) = \begin{cases} 1 & a = a^2 \\ 0 & a \neq a^2 \end{cases}$$

Therefore, we could say

$$\mathbb{E}_{\pi^1} \{R_{t+1} | S_t = s\} = 0 \quad \mathbb{E}_{\pi^2} \{R_{t+1} | S_t = s\} = 1$$

Playing in RL Framework: *Value Function*

Value of *state* s is

$$v_{\pi}(s) = \mathbb{E}\{G_t | S_t = s\} = \mathbb{E}\{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s\}$$

Transition function specifies *distribution of the next state* S_{t+1}

- Policy $\pi(a | S_{t+1})$ specifies *next action* A_{t+1}
- Rewarding function specifies the *next reward*

$$R_{t+2} = \mathcal{R}(S_{t+1}, A_{t+1})$$

↳ We can compute the second term ...

$$\mathbb{E}_{\pi}\{R_{t+2} | S_t = s\} \leftarrow \text{Expectation over } S_{t+1}, A_{t+1} \text{ and } A_t$$

Playing in RL Framework: *Value Function*

Value of state s is

$$v_{\pi}(s) = \mathbb{E}\{G_t | S_t = s\} = \mathbb{E}\{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s\}$$

So if we set the *policy* π , we could keep on

$$\mathbb{E}_{\pi}\{R_{t+3} | S_t = s\}, \mathbb{E}_{\pi}\{R_{t+4} | S_t = s\}, \dots$$

and we can average *future return* with *policy* π when we start at *state* s

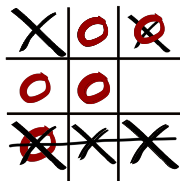
$$v_{\pi}(s) = \mathbb{E}_{\pi}\{R_{t+1} | S_t = s\} + \gamma \mathbb{E}_{\pi}\{R_{t+2} | S_t = s\} + \dots$$

Value Example: *Tic-Tac-Toe*

Tic-Tac-Tao Environment

We know that *opponent* takes the following strategy:

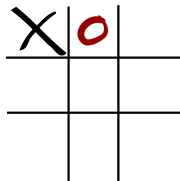
- if corners are available without chance of loosing, it plays a corner
 - ↳ it starts with cross-corner
 - ↳ it plays other corners afterwards
- if a line can be filled **by us** in next move, it blocks the line
 - ↳ if multiple lines are available, it blocks one of them **at random**
- if there is a line to be filled, it fills and wins



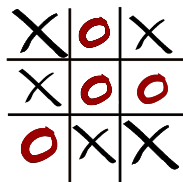
Value Example: Tic-Tac-Toe – Policy I

Tic-Tac-Tao Policy I

- if there is no chance of loosing, we play **next cell**
 - ↳ if next cell in the row available, we play that cell
 - ↳ if next cell in the row not available, we play cell below
 - ↳ if neither is available, we play some cell at random
- if there is a line possibly filled in next move by **opponent**, we block it
 - ↳ if multiple lines are available, we block one of them **at random**
- if there is a line to be filled by **us**, we fill and win



Value Example: Tic-Tac-Toe – Policy I



$$v_{\pi^1}(\{0\}) = 0$$

assuming $\gamma = 1$

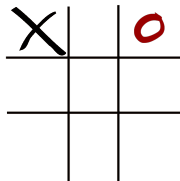
Say the game starts at **state** $s = \{0\}$ with **policy** $I: \pi^1$

- Our initial move will be $A_0 = 1$
- **Opponent** will play $O_1 = 8$
 $\hookrightarrow$ this leads to new state $S_1 = \{0, 1, 8\}$
- Our next move will be $A_1 = 4$
- **Opponent** will play $O_2 = 7$
 $\hookrightarrow$ this leads to new state $S_2 = \{0, 1, 4, 7, 8\}$
- Our next move will be $A_2 = 6$
- **Opponent** will play $O_3 = 2$
 $\hookrightarrow$ this leads to new state $S_3 = \{0, 1, 2, 4, 6, 7, 8\}$
- Our next move will be $A_3 = 5$
- **Opponent** will play $O_4 = 3$
 $\hookrightarrow$ this leads to new state $S_4 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

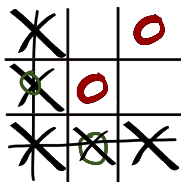
Value Example: Tic-Tac-Toe – Policy II

Tic-Tac-Tao Policy II

- if there is no chance of loosing, we play **a corner**
 - ↳ if a corner is available in the row, we play that corner
 - ↳ if no corner is available in the row, we play corner in the same column
 - ↳ if neither is available, we play some random cell
- if there is a line possibly be filled in next round by **opponent**, we block it
 - ↳ if multiple lines are available, we block one of them **at random**
- if there is a line to be filled by **us**, we fill and win



Value Example: Tic-Tac-Toe – Policy II



$$v_{\pi^2}(\{0\}) = -1$$

assuming $\gamma = 1$

Say the game starts at **state** $s = \{0\}$ with **policy** $I: \pi^2$

- Our next move will be $A_0 = 2$
- **Opponent** will play $O_1 = 8$
 - ↳ this leads to new state $S_1 = \{0, 2, 8\}$
- Our next move will be $A_1 = 4$
- **Opponent** will play $O_2 = 6$
 - ↳ this leads to new state $S_2 = \{0, 2, 4, 6, 8\}$
- Our next move will be **random**
 - ↳ with 50% chance we play $A_2 = 7$
 - ↳ **Opponent** will play $O_3 = 3$
 - ↳ with 50% chance we play $A_2 = 3$
 - ↳ **Opponent** will play $O_3 = 7$
- With any possible move, we end up **losing**

Playing in RL Framework: *Value Function*

Value function at a **state** can be also computed for *each possible next action*

Action-Value Function

Assume agent acts with policy $\pi(a|s)$ at **state** s ; then, action-value is

$$q_{\pi}(s, a) = \mathbb{E}_{\pi} \{G_t | S_t = s, A_t = a\}$$

Because it depends on **policy** π

Playing in RL Framework: Action-Value Function

Action-Value of *state-action* pair (s, a) is

$$q_{\pi}(s, a) = \mathbb{E}_{\pi} \{ R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s, A_t = a \}$$

Rewarding function specifies *distribution of* R_{t+1}

- For a given pair (s, a) , we get

$$p_{\ell}(s, a) = \Pr \{ R_{t+1} = r^{\ell} | S_t = s, A_t = a \}$$

- We can compute *expected immediate reward* as

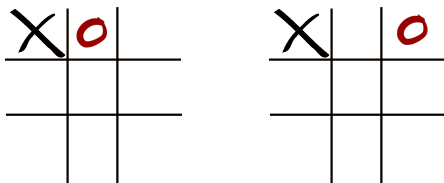
$$\mathbb{E} \{ R_{t+1} | S_t = s, A_t = a \} = \sum_{\ell=1}^L p_{\ell}(s, a) r^{\ell}$$

- We can transit to the next time ...

Action-Value Example: Tic-Tac-Toe – Hybrid Policy

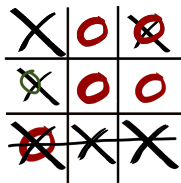
We play Tic-Tac-Toe with a *hybrid approach*

- with 50% chance we stick to *policy I*
- with 50% chance we stick to *policy II*



Let's denote this hybrid policy by $\bar{\pi}$

Action-Value Example: Tic-Tac-Toe – Hybrid Policy



Say the game starts at *state* $s = \{0\}$

- If we decide to stick to *policy I*
 - ↳ initial move will be $A_0 = 1$
 - ↳ we end up *drawing* $q_{\pi}(\{0\}, 1) = 0$
- If we decide to stick to *policy II*
 - ↳ next move will be $A_0 = 2$
 - ↳ we end up *losing* $q_{\pi}(\{0\}, 2) = -1$

Value vs Action-Value

The value and action-value functions are related as¹

$$v_{\pi}(s) = \sum_a \pi(a|s) q_{\pi}(s, a) = \mathbb{E}_{\pi} \{q_{\pi}(s, A)\}$$

¹Check Assignment 1

Playing in RL Framework: *Optimal Policy*

- + What is the application of *value functions*?
- They let us compare *policies*

Better Policy

We say policy π^1 is *better* than policy π^2 , and denote it by $\pi^1 \geq \pi^2$ if

$$v_{\pi^1}(s) \geq v_{\pi^2}(s)$$

for *all* $s \in \mathcal{S}$

Playing in RL Framework: Optimal Policy

It is important to note that when we say $\pi^1 \geq \pi^2$; it means that by playing π^1 , we expect *higher or equal* future return than π^2 , *no matter at which state we start using policy π^1*

This definition helps us defining **optimal policy**: simply speaking

optimal policy is the *policy* that is the *best*

Optimal Policy

Policy π^* is called *optimal* policy if for *any possible policy π* , we have

$$\pi^* \geq \pi$$

Playing in RL Framework: *Optimal Policy*

Optimal Policy: *Alternative Definition*

Policy π^* is called *optimal* policy if it maximizes the value for each *state*

$$\pi^* = \operatorname{argmax}_{\pi} v_{\pi}(s)$$

for *all* $s \in \mathcal{S}$

- + Can we guarantee that the *optimal policy* always exists?
- Yes!
- + Why didn't we define it based on *action-value function*?
- Well! We could, but we end up with the same result!

Playing in RL Framework: *Optimal Policy*

Basic Property of Optimal Policy

Let $\pi^\star$ be the *optimal* policy; then,

$$v_\star(s) = v_{\pi^\star}(s) = \max_{\pi} v_{\pi}(s)$$

for *all* $s \in \mathcal{S}$, and

$$q_\star(s, a) = q_{\pi^\star}(s, a) = \max_{\pi} q_{\pi}(s, a)$$

for *all* $s \in \mathcal{S}$ and $a \in \mathcal{A}$

Attention

Optimal policy is *not* necessarily unique: we may have multiple *optimal* policies

Ultimate Goal in RL: *Finding Optimal Policy*

Ultimate goal in an RL problem is to find the *optimal policy*

But there are a bunch of challenges in this way

① *How can we get a **model** of environment?*

↳ *Maybe, we could assume some model*

↳ *We assume some transition and rewarding functions*

↳ *Our RL framework is **based on a model***

↳ *Maybe, we could estimate what we need directly from enough samples*

↳ *We directly compute value or policy from samples, e.g.,*

$$q_{\pi}(s, a) = \mathbb{E} \{G_t | S_t = s, A_t = a\} \approx \text{average from samples}$$

↳ *Our RL framework is **free of a model***

High Level Classification of RL Methods

Model-Based RL

Bellman Equation

value iteration

policy iteration

Model-free RL

on-policy methods

temporal difference

Monte Carlo

SARSA

off-policy methods

Q-learning

Ultimate Goal in RL: *Finding Optimal Policy*

*Ultimate goal in an RL problem is to find the **optimal policy***

But there are a bunch of challenges in this way

② *How can we apply either approach in environments with **huge # of states**?*

↳ Maybe, we could use **neural networks**

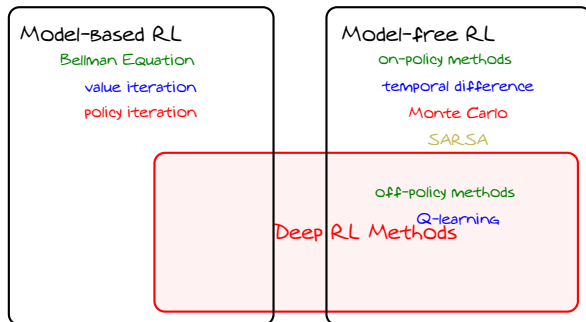
↳ We approximate the target function via a **neural network**, e.g.,

$$v_{\star}(s) \approx \text{CNN}(s|\mathbf{w})$$

↳ We collect some samples and train the network based on them

↳ We use that approximation instead of true function

High Level Classification of RL Methods



Episodic Environment

Terminal State

State s is called terminal if for any action $a \in \mathbb{A}$, we have

$$\mathcal{P}(s, a) = s \quad \text{and} \quad \mathcal{R}(s, a) = 0$$

A basic property of terminal state is that it has zero value: say s is terminal

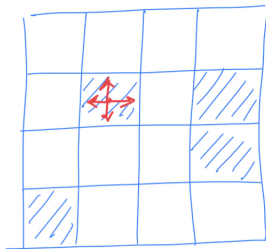
$$\begin{aligned} v_{\pi}(s) &= \mathbb{E}_{\pi} \{G_t | S_t = s\} = \mathbb{E}_{\pi} \{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots | S_t = s\} \\ &= \mathbb{E}_{\pi} \{0 + \gamma 0 + \gamma^2 0 + \dots | S_t = s\} = 0 \end{aligned}$$

Episode

The trajectory from starting state to a terminal state is often called an episode

Terminal State: *Frozen Lake*

- + How can we specify the end of the game?
- We can use the notion of **terminal state**



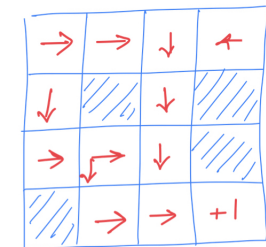
Once we lose, the **state** does not change anymore

$$\mathcal{P}(5, a) = 5$$

for all **actions** $a \in \mathbb{A} \rightsquigarrow s = 5$ is a terminal state

Frozen Lake: *Policy*

Policy in this problem describes the *planned path towards the goal*

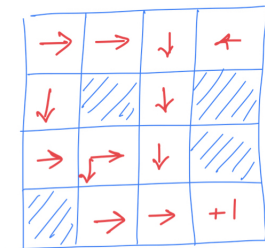


For instance in the above policy we have

$$\pi(a|1) = \begin{cases} 1 & a = 2 \\ 0 & a \neq 2 \end{cases} \quad \text{and} \quad \pi(a|9) = \begin{cases} 0.5 & a = 2 \\ 0.5 & a = 1 \\ 0 & a = 0, 3 \end{cases}$$

Frozen Lake: Value

Value describes the *chance of getting to the goal if we play the given policy*



For instance, if slipping probability is $p = 0$ the above policy leads to

$$v_{\pi}(s) = \begin{cases} 1 & s \text{ is not terminal} \\ 0 & s \text{ is terminal} \end{cases}$$

